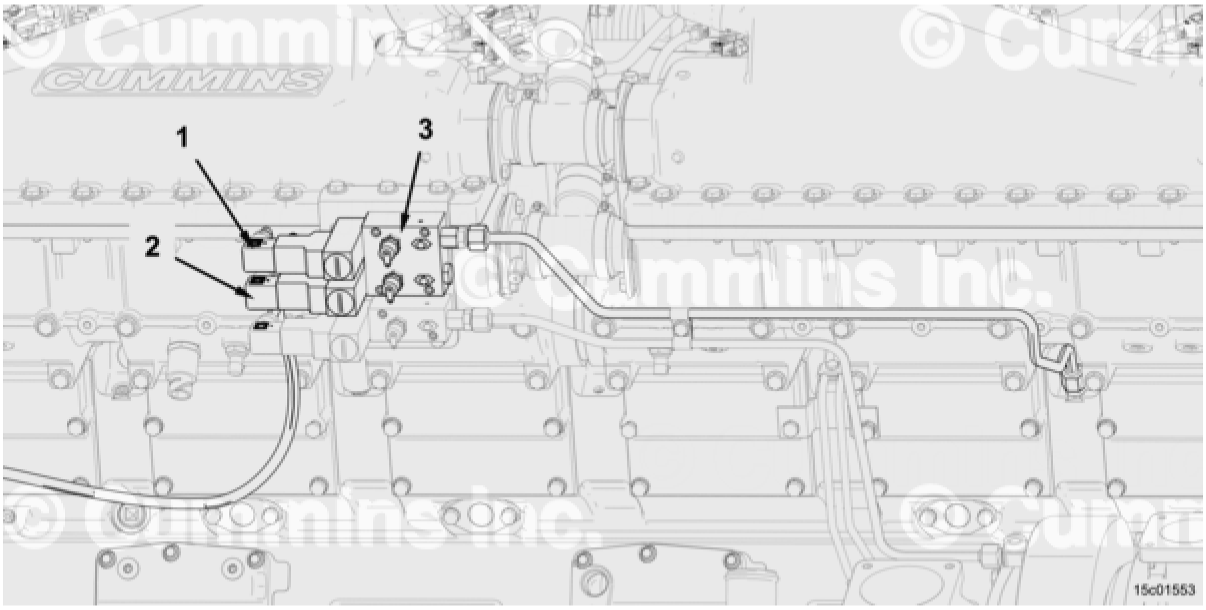
Trainings

Component Diagram



Alarm System Engine Lubricating Oil Pressure Switch Component Diagram

1. Alarm system low speed lubricating oil pressure switch
2. Alarm system high speed lubricating oil pressure switch
3. Switch manifold valves

Select Service Tools

Recommended Cummins® Service Tools

- Contact cleaner, Part Number 3824510, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent

Additional Service Items

- No additional service items required.

General Information

Two alarm system lubricating oil pressure switches are installed on the alarm system lubricating oil pressure switch manifold. The lubricating oil pressure switches measure lubricating oil pressure at the main oil rifle. The lubricating oil pressure switches are normally open switches that close when lubricating oil pressure falls below a set threshold. The low speed lubricating oil pressure switch is monitored by the customer interface box (CIB) when the engine is running. The high speed lubricating oil pressure switch is monitored by the CIB **only** when engine speed is above a set threshold.

The switches are connected to the CIB by the alarm system wiring harness. The connectors on the alarm system wiring harness are identical for the two switches. The mating connector on the wiring harness is a DIN 43650 connector.

The manifold is equipped with a valve to shut off flow to the pressure switch when the switch is being removed.

All components handled in this procedure weigh less than 23 kg [50 lb].

Preparatory Steps

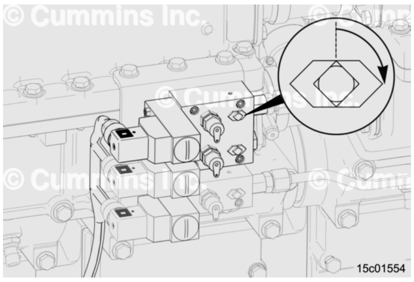
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect batteries and power supplies. See equipment manufacturer service information.
- Disconnect air supply line from air starting motor, if equipped. See equipment manufacturer service information.

Remove

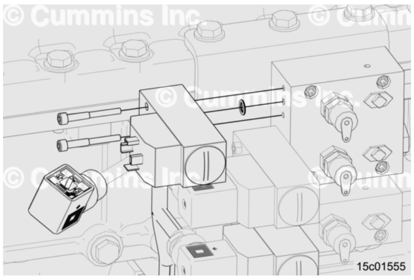
Close valve on switch manifold for switch being removed.



Disconnect wiring harness.

Remove switch.

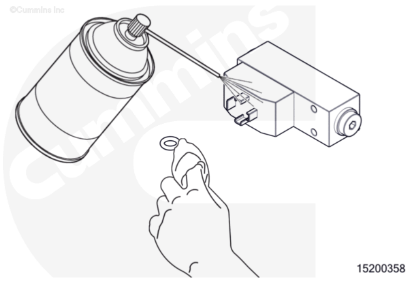
Remove o-ring seal. Save to inspect for reuse.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean switch. Use contact cleaner, Part Number 3824510, or equivalent.

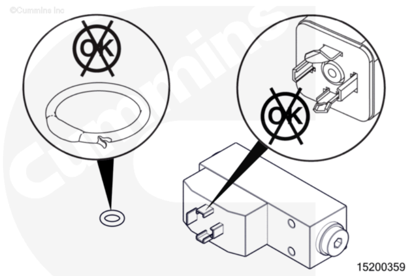
Clean o-ring seal. Use clean, lint-free cloth.

Inspect:

- Switch
- O-ring seal.

Replace switch if:

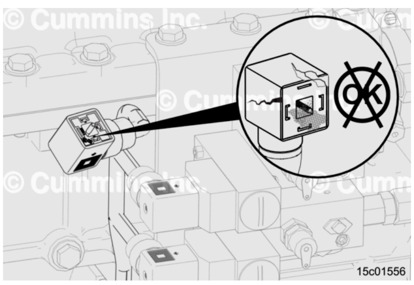
- Cracked
- Terminals damaged
- O-ring seal damaged
- Otherwise damaged.



Inspect wiring harness connector.

Replace connector if:

- Shell cracked or broken
- Seals missing or damaged
- Terminals contaminated with dirt, debris, or moisture



- Terminals corroded, bent, broken, pushed back, or expanded.

Install

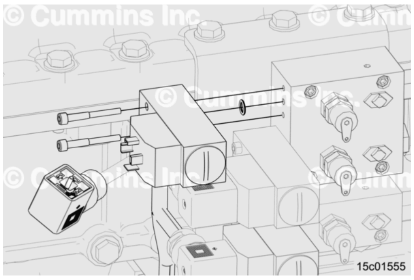
Apply small amount of Lubriplate™ 105, Part Number 3163086, to o-ring seal.

Install o-ring seal on switch.

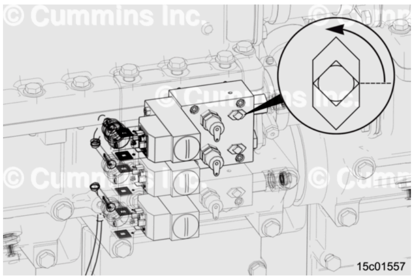
Install switch.

Torque Value: 8 n•m [71 in-lb]

Connect wiring harness.



Open valve on switch manifold.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect air supply line to air starting motor, if equipped. See equipment manufacturer service information.
- Connect batteries and power supplies. See equipment manufacturer service information.
- Operate engine. Check for leaks.